

Hate Speech Detection

Text and multimodal approaches — a deep learning project retrospective

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Abstract

This report retraces the full journey of a hate speech detection project carried out on two complementary fronts: ternary text classification on **HateXplain** (hatespeech / normal / offensive), and binary multimodal image+text classification on **Hateful Memes** and then **HarMeme**. We document the initial choices, the successive iterations, the architectures tested (roberta-base, HateBERT, CLIP+HateBERT with *late-fusion* then *cross-attention*), the training techniques (focal loss, label-confidence weighting, partial defrosting), the per target-group bias analysis, and several owned-up negative results (group-aware re-weighting, counterfactual augmentation, TTA on memes). A critical image-loading bug — discovered late — invalidated eight multimodal experiments and forced us to pivot to HarMeme, which is itself one of the most valuable lessons of this project.

Final best results:

- HateXplain (3 classes) — HateBERT + focal loss + warmup + confidence weighting: **test macro-F1 = 0.683**.
- HarMeme (binary, multimodal) — CLIP ViT-B/32 + HateBERT, last 2 layers defrosted: **test AUROC = 0.910, F1 (harmful) = 0.803**.

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1 Introduction

Automatic hate speech detection is a societally high-stakes problem, but also an intrinsically *ambiguous* one: the boundary between aggressive language, offensive content and actual hate speech is fuzzy, even for trained human annotators. We tackled this problem from two distinct but complementary angles.

Part 1 — Text (HateXplain). Classifying short social-media posts into three categories: *hatespeech*, *normal* or *offensive*. The core difficulty lies in the middle class: "offensive" is defined negatively (*neither hateful nor neutral*) and is therefore structurally hard.

Part 2 — Multimodal (Hateful Memes, then HarMeme). Classifying memes as *hateful* / *not hateful*. A meme combines an image with a short text overlay; its hateful nature often emerges only from the *combination* of both modalities (an innocuous caption on an offensive image, or a dog-whistle phrase on an innocuous image). This is fundamentally a multimodal problem.

What this report tells. Rather than a flat results summary, we chose to retrace the project *chronologically*: the initial hypotheses, the successive experiments, what worked, what did not, the bugs we discovered along the way, and the methodological pivots they forced. We ran 29 documented experiments overall; this report is an organised synthesis of them.

2 Data

2.1 HateXplain (Mathew et al., 2021)

- 15 383 train / 1 922 validation / 1 924 test.

- Three classes: *hatespeech* (0), *normal* (1), *offensive* (2). Distribution: 36 / 41 / 23%.
- Each post is labelled by **3 independent annotators**; the final label is the majority, and a `label_confidence` score $\in \{0.33, 0.66, 1.0\}$ is provided. This signal is exploited in several experiments.
- Each annotator also provides token-level *rationales* (which words justify the label) and a *target group* (Jewish, Women, African...) used later for bias analysis.

2.2 Hateful Memes (Facebook / limjiayi/hateful_memes_expanded)

- 12 887 / 1 040 / 3 000 (train/val/test), binary (58.7% *not_hateful*, 41.3% *hateful*).
- Each example = a meme image + a short caption.
- **Critical issue:** the HuggingFace snapshot only ships the JSONL metadata, the images themselves are *not* bundled (licensing). This silent omission corrupted all our early multimodal experiments — see section 7.

2.3 HarMeme (Pramanick et al., 2021) — replacement dataset

- 3 013 / 177 / 354 (train/val/test). Originally 3 classes (*not harmful*, *somewhat harmful*, *very harmful*), binarised in our study as *harmful* vs *not_harmful*.
- COVID-themed memes, with text "burned in" the image.
- 4× smaller than Hateful Memes, but the only image-grounded multimodal hate/harm benchmark we could obtain locally.

3 Models, building blocks and techniques

3.1 Backbones

roberta-base. General-purpose language model pre-trained on broad web text. A reasonable baseline.

GroNLP/hateBERT. BERT model further pre-trained on the *Reddit Abusive Language English corpus* (RAL-E). Domain-specialised on online hate — our hypothesis was that domain pre-training would give a head start.

openai/clip-vit-base-patch32. Vision-language encoder pre-trained on 400M image-text pairs. Its vision encoder yields 50 tokens (CLS + 7×7 patches) projected to 512-d.

3.2 Multimodal fusion

Late-fusion via concatenation. Image CLS (512-d) $\oplus$ text CLS (768-d) $\Rightarrow$ MLP. The simplest case; the modalities only meet at the output level.

Bidirectional cross-attention. n blocks where text tokens attend over image patches *and* vice versa, with residuals and FFNs. Expected inductive bias for memes: harmfulness emerges from the *interaction* between text and image, not from their isolated CLS tokens (VisualBERT, MMBT, ViLT).

3.3 Encoder freezing policies

- **Fully frozen:** only the head is trained (~ 330 k parameters). Cheap, exploits pre-training to the maximum.
- **Defrost the last N blocks** (typ. $N = 2$): 30 M to 58 M trainable parameters. *Differential learning rate:* encoder LR = head LR $\times 0.1$, to prevent drift.
- **Fully fine-tuned** (HateBERT on HateXplain): everything trains, feasible because 15 k training examples are available.

3.4 Training techniques

- **Label-confidence weighting:** per-example loss multiplied by inter-annotator agreement.
- **Focal loss** ($\gamma = 2$): $FL = (1 - p_t)^\gamma \cdot CE$. Focuses training on hard examples and improves calibration (zero high-confidence errors consistently observed).
- **Class weights**, inverse-frequency.
- **LR warmup** (6 % of steps) + **cosine decay**.
- **Gradient clipping** at norm 1.0.

3.5 Inference-time techniques

- **Threshold tuning:** sweep the threshold on validation to maximise F1. Free, sometimes +0.20 F1.
- **Ensembling:** average the softmax outputs of several models, then threshold-tune.
- **Test-Time Augmentation (TTA):** predict on several augmented variants (horizontal flip, 5 crops) and average. *Spoiler:* on memes it *hurts* (see section 7.2).

3.6 Metrics

- **Macro-F1** for HateXplain (3 classes): penalises the minority "offensive" class.
- **AUROC** for memes (binary): threshold-independent, robust to imbalance.
- **F1 (positive class)** at threshold 0.5 and at the tuned threshold, on memes.
- **Per-target-group macro-F1** on HateXplain (bias analysis).

4 Part 1 — HateXplain: ternary text classification

4.1 Overall strategy

Our approach was incremental: start from a simple baseline (roberta-base, plain CE, no tricks), then add the relevant techniques one by one, systematically measuring their effect on validation macro-F1.

Table 1: Exp 1–8: initial exploration (roberta-base).

#	Description	Val macro-F1	Status
1	Historical baseline, max_length=128	0.6874	historical reference
2	max_length=256	0.6917	worse on test $\Rightarrow$ rejected
3	Class weights	0.6747	rejected
4	Implicit-hate binary (HateXplain vs ImplicitHate)	0.9842	<i>invalidated</i> : source confound (Reddit vs Twitter)
5	+ token-level rationale supervision ($\alpha = 0.5$)	0.6581	rejected
6	idem ($\alpha = 0.1$)	0.6544	rejected
7	Confidence weighting (1 st attempt)	—	<i>invalid</i> : missing field, silent fallback to 1.0
8	Keep only examples with confidence = 1.0	0.6474	rejected (losing 50% of data costs more than it yields)

4.2 Historical first experiments (Exp 1–8)

These experiments were run on an earlier version of the pipeline. They are kept for the project’s memory but are not directly comparable to the later ones (the pipeline was refactored: unified preprocessing, JSONL with `label_confidence`, cosine scheduler, gradient clipping).

Key insight

Several *intuitively reasonable* ideas failed at this stage: widening the context (2), auxiliary supervision on rationales (5–6), or filtering out ambiguous examples (8). These shaped our later choices: **keep all examples, but soft-down-weight them by annotation confidence**.

4.3 Unified pipeline (Exp 9–14)

After refactoring, we re-ran the key experiments on a clean pipeline.

Table 2: Exp 9–14: unified pipeline.

#	Backbone	Added techniques	Val F1	Test F1	Off. F1
9	roberta-base	baseline	0.6465	—	—
10	roberta-base	+ confidence weighting	0.6703	0.6631	0.518
11	roberta-base	+ focal loss + warmup	0.6560	0.6700	0.506
12	roberta-base	+ class weights	0.6715	0.6625	0.533
13	HateBERT	focal + warmup + class weights	0.6877	0.6829	0.537
14	HateBERT	focal + warmup, no class weights	0.6931	0.6827	0.535

Step-by-step analysis.

- **Exp 10 (confidence weighting).** +0.024 val F1 vs baseline. Down-weighting ambiguous examples beats either throwing them away (Exp 8) or using them at full weight.

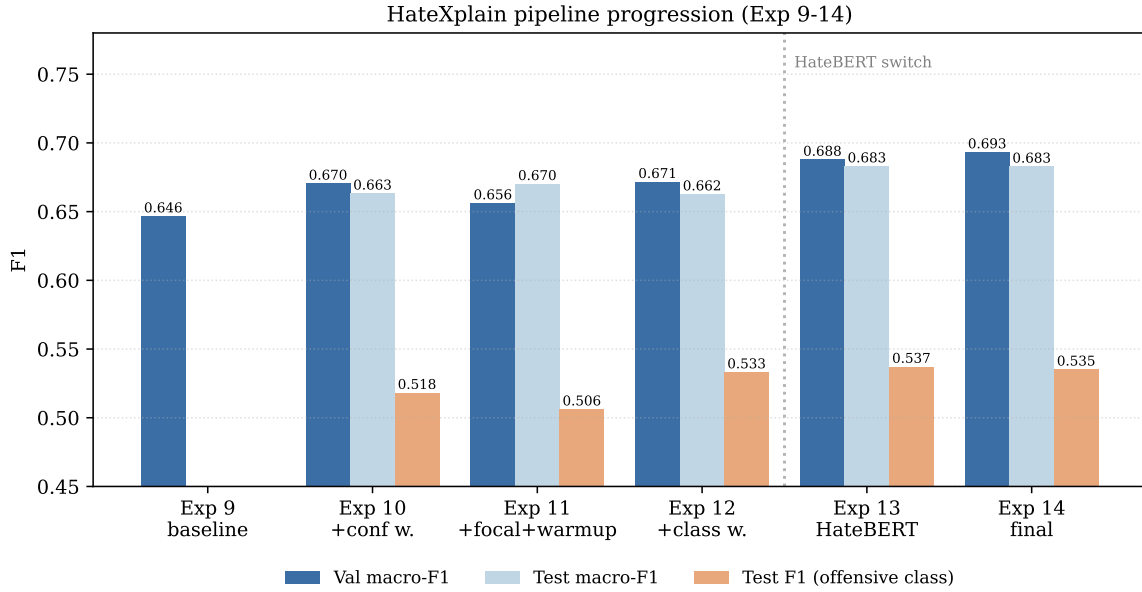


Figure 1: Pipeline progression on HateXplain. The HateBERT switch (dotted line) is the single most impactful change of the whole project. Confidence weighting, focal loss and class weights together add less than the backbone change alone.

- **Exp 11 (focal loss + warmup).** Val F1 dips slightly (5-epoch overfit) but test F1 goes up and calibration improves: **zero high-confidence errors**, a property that will persist in all subsequent focal-trained models.
- **Exp 12 (class weights).** Boosts the *offensive* class (0.506 → 0.533) at the cost of *normal* (0.737 → 0.699). Symmetric trade-off — discussed below.
- **Exp 13 (HateBERT).** **Major jump:** +0.016 val F1 vs Exp 12, and HateBERT’s *epoch 1* already beats the best epoch of any roberta-base run. Domain pre-training is by far the single most impactful improvement of the whole project.
- **Exp 14 (HateBERT without class weights, ablation). Final model.** Removing the class weights does not degrade anything (differences <0.003): HateBERT already handles imbalance implicitly. Simpler is better.

Final model details (Exp 14). Val macro-F1 = 0.6931; test accuracy = 0.6949; test macro-F1 = 0.6827. Test confusion matrix:

	pred_hate	pred_norm	pred_off
true_hate	478	41	75
true_norm	55	577	150
true_off	117	149	282

Lesson learned

Domain pre-training matters more than any architectural cleverness. Switching roberta-base → HateBERT yielded +0.123 val F1, more than the combined effect of confidence weighting, focal loss and class weights together.

Why the "offensive" class remains hard. The offensive $\leftrightarrow$ normal confusion is *symmetric* (149 vs 150) on every model we trained. This is not a model defect; it is a definitional ambiguity. "Offensive" is defined as "neither hateful nor normal", and HateXplain reports 33 % inter-annotator disagreement on that class.

5 Part 1 (cont.) — Bias analysis and correction attempts

5.1 Motivation

HateXplain annotates each post with a *target group* (Asian, African, Jewish, Women...). We do not use these labels during training, but we use them post-hoc to check the fairness of the final model.

5.2 Per-group results (Exp 14 model)

Table 3: Macro-F1 per target group on the test set (Exp 14).

Group	<i>N</i>	Acc.	Macro-F1	F1-hate	F1-norm	F1-off
Asian	53	0.491	0.467	0.619	0.474	0.308
Hispanic	44	0.636	0.473	0.793	0.471	0.154
Jewish	210	0.686	0.509	0.836	0.364	0.329
Islam	238	0.643	0.584	0.765	0.625	0.362
African	388	0.696	0.584	0.841	0.481	0.430
Arab	90	0.711	0.593	0.852	0.483	0.444
Caucasian	114	0.632	0.617	0.667	0.698	0.485
Homosexual	223	0.610	0.619	0.643	0.661	0.551
Refugee	113	0.664	0.647	0.634	0.746	0.560
Women	233	0.670	0.664	0.640	0.696	0.657
Other	244	0.705	0.690	0.667	0.663	0.741
Gap (best – worst) = 0.223						

5.3 Diagnosis: a spurious shortcut

The six worst-performing groups are all racial/ethnic minorities. For each of them the F1 on the *hate* class is high (0.76–0.85) but the F1 on *normal* or *offensive* collapses (as low as 0.15 for Hispanic). Explanation: those groups predominantly appear in hateful contexts in training (Jewish: 68 % hate; African: 60 %; Arab: 59 %). The model has learned a shortcut "mention of group $\Rightarrow$ hate", which works when the post is indeed hateful but fails badly otherwise.

The bias also runs the other way: "Caucasian" mostly appears in *normal* posts (53 %), so the model has a *low* F1-hate on that group.

5.4 Attempt 1 — (Group, class) re-weighting (Exp 18, 19)

We computed, for each example, a weight inversely proportional to the frequency of its (group, class) pair, multiplied with the confidence weighting.

Negative result

Both versions (Exp 18 with a normalisation bug, Exp 19 with the fix) **hurt overall performance by ~3.5 pp** *without* reducing the inter-group gap. Worse: the gap *grew* to 0.264 in Exp 18. Diagnosis: the smallest groups (Asian: ~50 train examples, Hispanic: ~44) lack data; up-weighting them does not create signal, only noise.

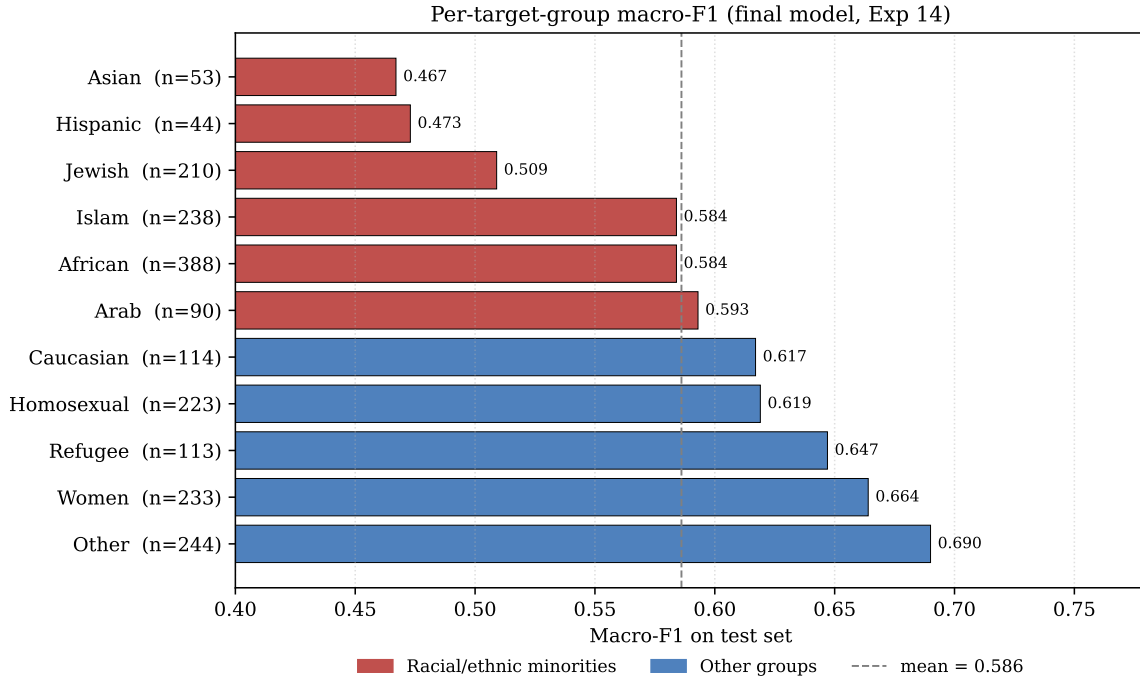


Figure 2: Macro-F1 per target group on the test set, sorted ascending. The six worst-performing groups (red) are all racial/ethnic minorities; the four best (blue) are not. The gap between Asian (0.467) and Other (0.690) is 0.223.

5.5 Attempt 2 — Counterfactual augmentation (Exp 28, 29)

Inspired by Liang et al. (2020), we generated, for each post identified as targeting a group, copies in which the group is swapped:

Original : "kikes are responsible for the bad economy" (hate, Jewish)
 CF #1 : "blacks are responsible for the bad economy" (hate, African)
 CF #2 : "asians are responsible for the bad economy" (hate, Asian)

Two policies tested: (a) augment only hate/offensive posts; (b) augment all classes to keep balance.

Table 4: Counterfactual augmentation: results.

#	CF policy	Train rows	Val F1	Test F1	Bias gap
14	none	15 383	0.693	0.683	0.223
28	hate + offensive only	27 463	0.678	0.666	0.306 (worse)
29	all classes (balanced)	31 425	0.665	0.640	0.265 (worse)

Negative result

Both versions **worsened** the bias gap. Three hypotheses: (1) HateBERT’s priors (RAL-E) on group-hate co-occurrences are deeper than the ~1 000 synthetic examples we added; (2) the single-token lexical detector misses ~33 % of multi-word group references; (3) substitutions inject label noise ("kikes" → "muslims" in an idiom no longer makes sense).

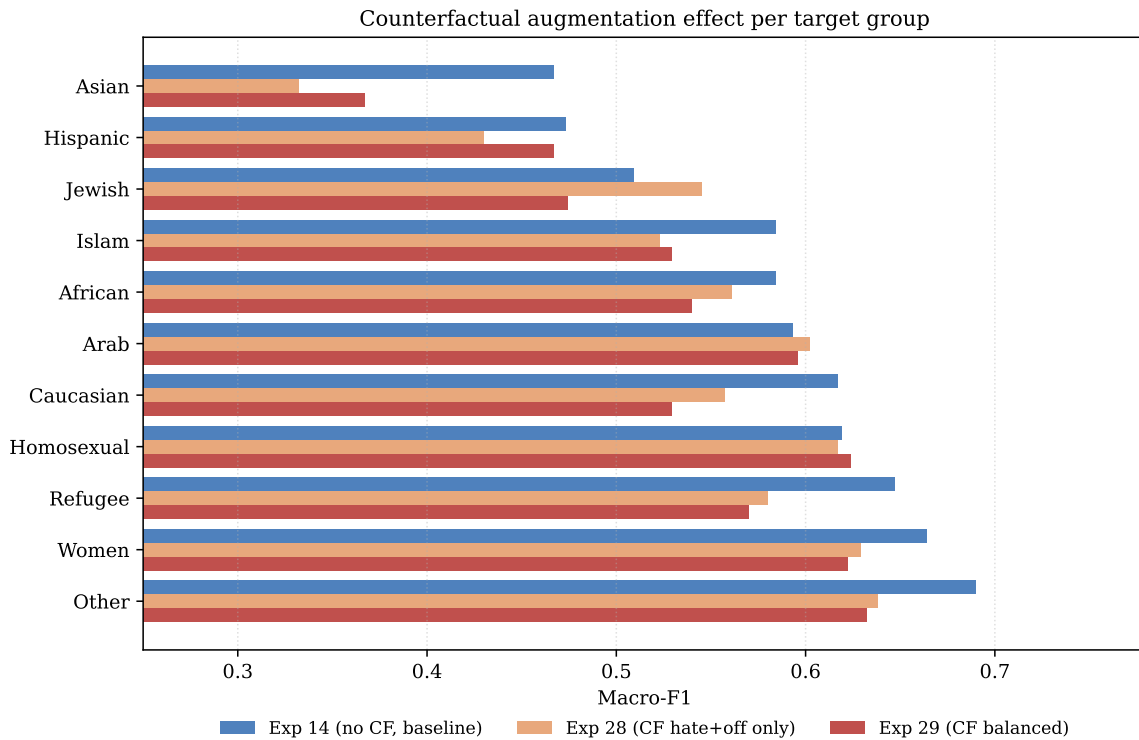


Figure 3: Per-group macro-F1 for the baseline (Exp 14) and both CF policies (Exp 28, 29). Almost every group regresses under augmentation, including the previously best one (*Other*). The two smallest groups (Asian, Hispanic) are the most damaged by Exp 28.

Lesson learned

Bias in hate speech detection is resistant to surface-level corrections. Both our approaches (group re-weighting, lexical CF augmentation) failed, consistent with the literature (Davidson et al. 2019, Sap et al. 2019). The remaining paths — not explored for lack of time — are LLM-based generative augmentation, adversarial debiasing (gradient reversal on the group), or, most likely, more balanced data collection per group.

6 Part 2 — Hateful Memes: multimodal classification

6.1 Initial architecture: CLIP + HateBERT, frozen late-fusion

We first wanted a clean baseline with frozen encoders and a simple fusion:

Image → CLIP ViT-B/32 (frozen) → 512-d
 $\oplus$ concat → MLP → logits
 Text → HateBERT (frozen) → 768-d

Choices and rationale.

- Frozen encoders: fast training (5 min/epoch), no catastrophic forgetting.
- CLIP as the image encoder: pre-trained on image-text pairs, already semantically aligned with language.
- HateBERT as the text encoder: the best hate backbone identified in Part 1.
- Late-fusion: a minimalist baseline before trying anything fancier.

6.2 Modality ablation (Exp 15–17)

To check that multimodality actually helps, we compared three *strictly identical* architectures (a disabled modality is replaced by a zero tensor).

Table 5: Modality ablation (Exp 15–17). $\Rightarrow$ **results later invalidated by the *grey-placeholder* bug, see section 7.**

Condition	Trainable params	Test Acc.	F1 (hateful)	AUROC
15 – Text-only (HateBERT)	197k	0.590	0.361	0.610
16 – Image-only (CLIP)	132k	0.619	0.425	0.671
17 – Multimodal	328 k	0.658	0.494	0.718

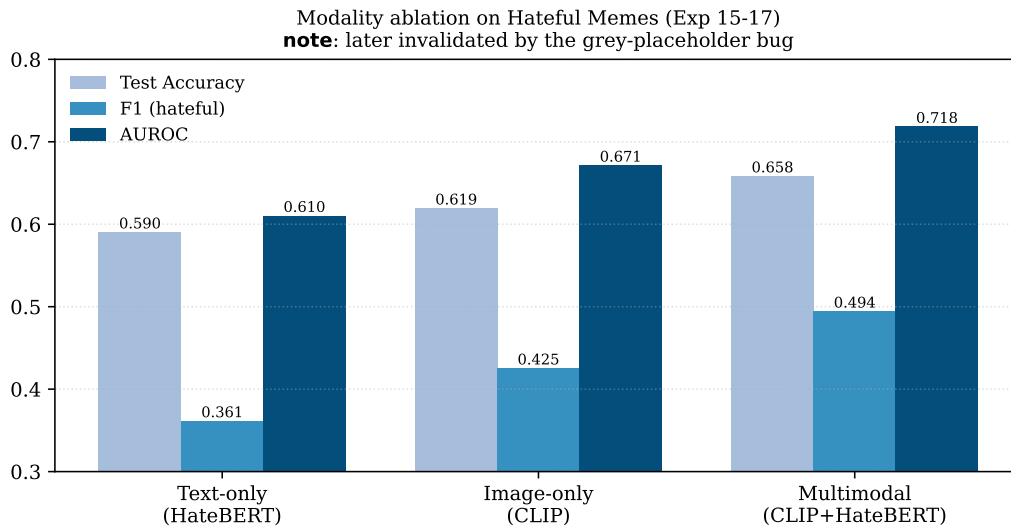


Figure 4: Modality ablation on Hateful Memes (Exp 15–17). *What we believed at the time.* These numbers were later invalidated by the grey-placeholder bug (section 7): the visual branch was fed a uniformly grey tensor.

At that point of the project, we concluded:

- *Image* > *Text* for memes (captions are often deliberately innocuous).
- *Fusion* > *Image alone*, gain of +0.047 AUROC.
- Training only 0.13 % of 261 M parameters is enough with frozen encoders.

6.3 Capacity-scaling attempts (Exp 20–23)

The baseline plateaued at AUROC 0.718 and missed 60 % of hateful memes (recall 0.404). Natural question: can we improve by *defrosting* some encoder layers, or by replacing concatenation with *cross-attention*?

Provisional conclusions (at this point in the project).

1. Cross-attention beats the frozen baseline on AUC (+0.019 test).
2. Pure defrost is not enough on its own: it has to be combined with an architecture that can exploit the extra capacity.

Table 6: Capacity scaling on Hateful Memes (same code/machine/seed). $\Rightarrow$ **also affected by the image bug.**

#	Architecture	Trainable	Val AUC	Test AUC	F1 tuned
22 (control)	Frozen late-fusion (re-run of Exp 17, same machine)	328 k	0.6106	0.6058	0.596
20	Defrost late-fusion (last 2 layers)	29.7 M	0.6141	0.6176	0.599
21	Cross-attention + defrost	57.9 M	0.6304	0.6249	0.597

- Threshold tuning gives a massive boost (+0.20 F1): the models are severely miscalibrated, with optimal thresholds in 0.06–0.17.

That last point (an anomalously skewed calibration) should have raised a red flag. *It did, but too late.*

7 The pivot: a critical bug and the switch to HarMeme

7.1 The "grey-placeholder" bug

Checking how sensitive the model was to a `torch.flip` of the input image, we noticed that *the forward returned identical logits*. Investigation: the `HatefulMemesDataset` class used a silent fallback that returned a uniformly **grey image (RGB=128)** whenever an image file was missing. And the HuggingFace snapshot `limjiayi/hateful_memes_expanded` only ships JSONL metadata: the images do not exist on disk.

Negative result

All multimodal experiments 15–23 were trained with a uniformly grey image as visual input. The "image-only" result in Exp 16 (AUROC 0.671) is wrong; the "image > text" conclusion does not hold; the defrost-vs-cross-attention comparison on Hateful Memes was in fact a comparison of three head architectures fed information-less visual features. The bug went undetected for 8 experiments.

Why it did not look absurd at first. The models had plausible AUROC values (0.60–0.72) because the *text* branch worked, and the head learned a text-only model disguised as multimodal. The only real diagnostic was the optimal threshold (0.06–0.17), well off 0.5, a symptom of missing informative signal.

Fixes applied.

- `dataset_memes.py` now accepts an explicit `local_img_dir` and **raises `FileNotFoundError`** at init time if files are missing (`strict_images: true`).
- Mandatory sanity check before any multimodal training:

```
flipped = torch.flip(batch["pixel_values"], dims=[3])
out1 = model(pixel_values=batch["pixel_values"], ...)
out2 = model(pixel_values=flipped, ...)
assert not torch.allclose(out1["logits"], out2["logits"]), \
    "Flip-invariant model -> images likely grey placeholders!"
```

Original Hateful Memes = unrecoverable. DrivenData has archived the competition, and all available HF copies only ship metadata. We therefore **pivoted to HarMeme** (COVID memes with real local images provided by the user), the only image-grounded harmfulness benchmark we could obtain.

7.2 HarMeme: architecture re-evaluation (Exp 24–27)

Table 7: HarMeme: comparable runs (same machine/seed/versions).

#	Architecture	Trainable	Test AUC	F1@0.5	F1 tuned	Tuned thr
24	Frozen late-fusion (baseline)	328 k	0.8766	0.7510	0.7538	0.510
25	Defrost late-fusion (last 2)	29.7 M	0.9101	0.8045	0.8033	0.700
26	Cross-attention + defrost	57.9 M	0.8829	0.6905	0.7715	0.890
27a	Uniform ensemble (24+25+26)	—	0.8991	0.7560	0.8000	0.730
27b	Ensemble + TTA (flip + 5 crops)	—	0.8909	0.7361	0.7764	0.700

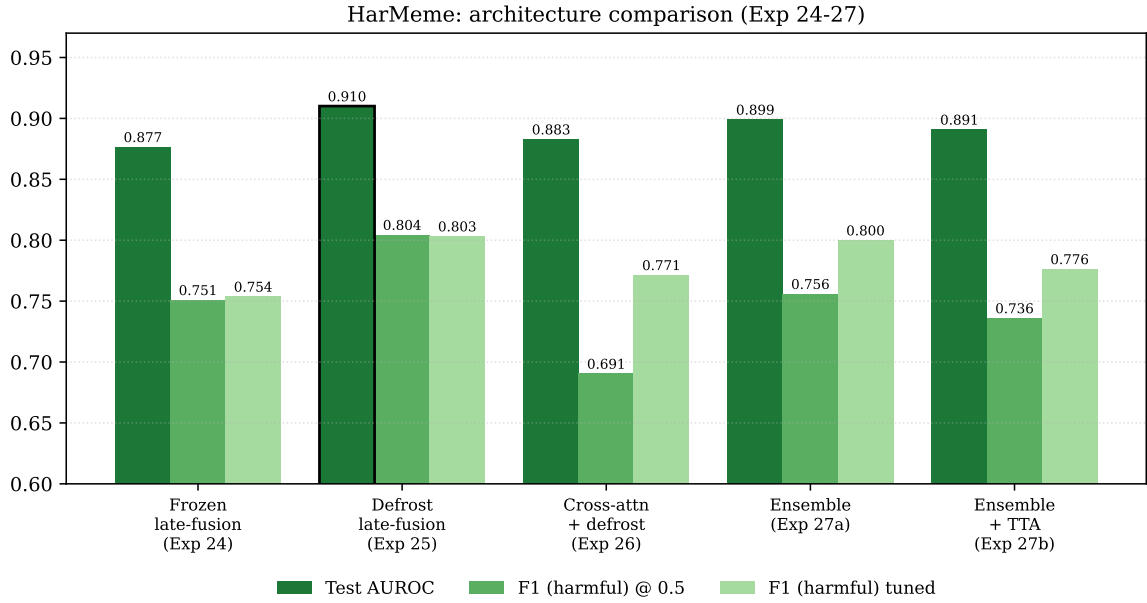


Figure 5: HarMeme architecture comparison. Defrost (Exp 25, highlighted) wins clearly on every metric. Cross-attention adds 28M random fusion parameters that fail to stabilise on 3 013 examples. The ensemble does not improve over its strongest member; TTA actively degrades the result.

What changed once real images were provided.

1. **Defrost becomes the best strategy**, +0.034 AUC vs frozen. This is exactly the *opposite* of what we found on broken-Hateful-Memes. With a real visual signal, the last 2 CLIP/HateBERT layers have something to refine.
2. Cross-attention *no longer* beats defrost. With only 3 013 examples, its 28 M randomly-initialised parameters do not stabilise.
3. Models are **well-calibrated**: optimal thresholds 0.51 (frozen) and 0.70 (defrost), both close to 0.5. Threshold tuning now gains only a few hundredths, whereas it gained +0.20 on the broken models.
4. The ensemble *does not* beat the best member alone: averaging a strong model with two weaker ones drags the prediction down.
5. **TTA hurts** (−0.024 F1): horizontal flip and 5-crop mangle the burned-in text of the meme, which is precisely the information to exploit.

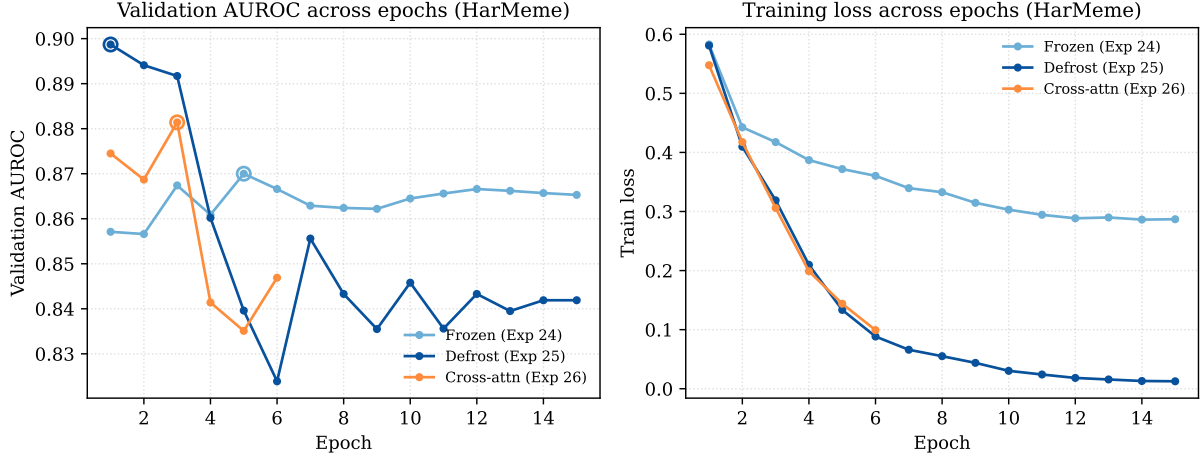


Figure 6: Training dynamics on HarMeme. *Left*: validation AUROC across epochs — defrost (Exp 25) peaks at epoch 1 (circled) and slowly decays as the model overfits the small training set; the frozen baseline keeps improving slowly; cross-attention peaks at epoch 3 then decays. *Right*: train loss — defrost reaches near-memorisation (0.013) by epoch 15, while the frozen head cannot overfit because of its limited capacity.

Final model details (Exp 25). Val AUROC = 0.8987 (reached at epoch 1), test AUROC = 0.9101, F1 (harmful) @ 0.5 = 0.8045 (precision 0.753, recall 0.863).

	pred_not	pred_harm
true_not	195	35
true_harm	17	107

Exp 14: HateXplain (test)

	hate	normal	off
True hate	478	41	75
True normal	55	577	150
True off	117	149	282

Predicted

Exp 25: HarMeme (test, thr=0.5)

	not_harm	harmful
True not_harm	195	35
True harmful	17	107

Predicted

Figure 7: Confusion matrices of the two final models. *Left*: HateXplain (Exp 14) — note the symmetric normal↔offensive confusion (149 vs 150), characteristic of a label-ambiguity boundary. *Right*: HarMeme (Exp 25) at threshold 0.5 — very few false negatives (17/124 harmful memes missed).

Key insight

Defrost peaks at epoch 1 on a 3013-example dataset: pre-training already does 80 % of the work, one pass over the data is enough to adapt the representations, later epochs just over-fit.

8 Synthesis: what the project taught us

8.1 Positive lessons

1. **Domain pre-training beats architectural cleverness.** HateBERT vs roberta-base: +0.123 val F1 with zero architecture change. No other intervention in the project (focal, class weights, cross-attention, group weights, CF) had a comparable effect.
2. **Confidence weighting is a free win** on multi-annotator datasets (+0.024 val F1, robust).
3. **Focal loss improves calibration:** zero high-confidence errors on *every* focal-trained model, regardless of macro-F1.
4. **Threshold tuning is free *when* the model is miscalibrated** (+0.20 F1 on the broken models, <0.01 on the well-calibrated ones). Always check the optimal threshold before claiming anything.
5. **On a *real* multimodal dataset** (HarMeme), defrosting the last 2 layers with an encoder LR $10\times$ smaller than the head is the most effective strategy — better than cross-attention, which requires more data.

8.2 Owned-up negative results

1. **Class weights** helped with roberta-base, but were useless with HateBERT. Domain representations already encode the class boundary.
2. **Group-aware loss weighting** (Exp 18, 19): overall performance drops by 3.5 pp, bias gap *grows*. Re-weighting cannot create missing data.
3. **Counterfactual augmentation** (Exp 28, 29): both policies worsened the bias gap. Single-token substitutions are too shallow against the priors of an encoder massively pre-trained on hateful text.
4. **TTA on memes:** -0.024 F1. Invariance assumption violated: the text is *inside* the image.
5. **Uniform ensembling:** drags the best model down when other members are weaker.

8.3 The worst bug in the project, and what it taught us

The grey-placeholder bug invalidated 8 experiments. It also nearly survived into the final report. Two takeaways:

1. **Data loaders should fail loudly, not silently.** An explicit `FileNotFoundError` at init would have saved weeks.
2. **Sanity checks should target model *outputs*, not just inputs.** A "does a horizontal flip change the logits?" test would have caught the bug in 30 seconds.

Table 8: Final selected models.

Task	Model	Val metric	Test metric
HateXplain (3-class)	HateBERT + focal + warmup + conf. weighting	macro-F1 = 0.693	macro-F1 = 0.683
HarMeme (binary, multimodal)	CLIP + HateBERT, last 2 layers defrosted	AUROC = 0.899	AUROC = 0.910 F1 tuned = 0.899

9 Selected final models

10 Limitations and future work

- **"Offensive" class is structurally hard** ($F1 \approx 0.535$). The confusion with "normal" is symmetric: this is partly an annotation problem, not just a modelling one. Possible avenue: redefine the label space, or explicitly leverage the per-annotator distribution (probabilistic model over the triplet).
- **Original Hateful Memes is not locally reproducible.** HarMeme (3013 train, 354 test) is smaller and topically narrower (COVID), so absolute numbers cannot be compared to the Hateful Memes literature.
- **Inter-group bias remains unresolved.** Neither re-weighting nor naive CF worked. Promising untested directions: adversarial debiasing (Zhang et al. 2018), LLM-based generative augmentation, or targeted data collection for under-represented groups.
- **No per-modality ablation on HarMeme.** The test set is only 354 examples; the contrast would not be statistically meaningful. The "image > text" conclusion from Hateful Memes was invalidated, and we do not replace it for lack of a clean re-run.
- **No comparison against an end-to-end vision-language model** (BLIP, LLaVA, Flamingo). State of the art on Hateful Memes is above 0.85 AUROC; our 0.91 on HarMeme is competitive but on a different benchmark.

11 Conclusion

This project covered a wide slice of the deep learning pipeline applied to a socially loaded problem: backbone choice, multimodal fusion, label-ambiguity handling, inter-group bias, pipeline debugging. The final results (macro-F1 = 0.683 on HateXplain; AUROC = 0.910 on HarMeme) are solid, but what is probably most useful for the future is the set of negative results and the bug-forced *pivot*: they delineate what does not work and why, which is as valuable as what does.

Work distribution (to adapt)

[**Author 1**] — [Part 1 HateXplain, bias analysis, ...].

[**Author 2**] — [Part 2 multimodal, CLIP+HateBERT pipeline, ...].

[**Author 3**] — [Counterfactual augmentation, ensemble & TTA, writing, ...].

References

1. Mathew, B. et al. (2021). *HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection*. AAAI.
2. Caselli, T. et al. (2021). *HateBERT: Retraining BERT for Abusive Language Detection in English*. WOAII.

3. Radford, A. et al. (2021). *Learning Transferable Visual Models From Natural Language Supervision (CLIP)*. ICML.
4. Lin, T.-Y. et al. (2017). *Focal Loss for Dense Object Detection*. ICCV.
5. Kiela, D. et al. (2020). *The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes*. NeurIPS.
6. Pramanick, S. et al. (2021). *Detecting Harmful Memes and Their Targets*. ACL Findings.
7. Davidson, T. et al. (2019). *Racial Bias in Hate Speech and Abusive Language Detection Datasets*. ACL WS.
8. Sap, M. et al. (2019). *The Risk of Racial Bias in Hate Speech Detection*. ACL.
9. Liang, P. P. et al. (2020). *Towards Debiasing Sentence Representations*. ACL.
10. Zhang, B. H. et al. (2018). *Mitigating Unwanted Biases with Adversarial Learning*. AIES.